MIL-A-45646(Ord)
10 May 1961
SUPERSEDING
FA(FC)-PD-537
21 August 1959

MILITARY SPECIFICATION

AMPLIFIER, ELECTRONIC CONTROL, NULLMETER: 8644119

1. SCOPE

1.1 This specification covers one type of nullmeter amplifier used in Gun Direction Computers: M15 and M15C, which are part of Field Artillery Fire Control Systems: M35 and M35C. The nullmeter amplifier, hereinafter referred to as "amplifier", is used as a phase sensitive device when driving a 100-0-100 microamperes direct current ammeter.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment, and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1220

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DRAWINGS

Ordnance Corps

F8643974	Electrical Schematic, Nullmeter Amplifier
F8644119	Nullmeter Amplifier

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one amplifier, representative of production, shall be made available to the Government to determine that the amplifier meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The amplifier shall be manufactured in accordance with Drawing F8644119, "Nullmeter Amplifier", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3.1 Ground potential.- The design and construction of the amplifier shall insure that all external parts, exclusive of electrical connections, are at ground potential at all times.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

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3.5 Environmental.

3.5.1 Vibration.- The amplifier shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.03 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the amplifier shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.2 Operating temperatures.- The amplifier shall meet the requirements of 3.6 and 3.7 when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.5.3 Storage temperatures.- The amplifier shall show no evidence of physical failure and shall meet the requirements of 3.6 and 3.7 at room temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 65 degrees Fahrenheit.

3.6 Circuitry.- Circuitry shall be in accordance with electrical schematic, Drawing F8643974.

3.7 Performance.- An externally mounted Ammeter, 8643039 shall be considered integral with the amplifier.

3.7.1 Operating voltages.- The amplifier shall be capable of operating from plus 28 volts plus or minus 0.1 percent, dc regulated and 2.5 volts (nominal), 400 cps, phased instantaneously positive and negative as referred to a common neutral.

3.7.2 Output signal.- The output signal shall appear as a dc voltage across the specified meter.

3.7.2.1 When the input voltage is in phase with the reference voltage, the dc output voltage shall be positive.

3.7.3 Sensitivity.- The gain of the amplifier shall be such that a 20-millivolt plus or minus 10 percent input signal shall produce a full scale deflection on the meter of plus or minus 100 microamperes.

3.7.3.1 The amplifier shall saturate so that not more than 0.8 milliampere, dc, shall appear across the meter.

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3.8 Alinement and calibration.- Alinement and calibration of all amplifiers shall be performed using the same power source to insure uniformity of product. The reference voltages supplied, shall be from a power source maintained as a standard.

3.9 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions.- The quality assurance provisions of referenced documents, to the extent specified herein, together with provisions of this section are applicable to inspection performed by the contractor and for Government verification.

4.1.1 The contractor shall provide an inspection system in accordance with Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one amplifier shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements affecting concentricity, squareness, parallelism,

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micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the amplifier. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.4.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sampling Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.4.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.5 Material.- Control of material shall be in accordance with Test Data requirements of Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

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4.6.2 Accuracy requirements for special testing devices.

4.6.2.1 Device to test vibration requirements.- The device shall consist of a vibrating machine which will produce motion as specified. It shall be equipped with a medium capable of holding the unit securely to the vibrating machine in the specified positions.

4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None defined

MAJOR:	AQL 4.0%	Req't Par.	Q.A. Par.	Test Method
101.	Post vibration	3.5.1	4.7.2	Device, Special Testing (DST)
102.	Operating temperature	3.5.2	4.7.3	DST

MINOR: None defined

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None defined

MAJOR:	AQL 1.5%	Req't Par.	Q.A. Par.	Test Method
101.	Operating voltages	3.7.1	4.7.5	Stndrd Meas Equip (SME)
102.	Output signal	3.7.2	4.7.6	SME

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TABLE III - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 1.5%		Req't Par.	Q.A. Par.	Test Method
103.	Output phase	3.7.2.1	4.7.7	SME
104.	Sensitivity	3.7.3	4.7.8	SME
105.	Saturation	3.7.3.1	4.7.9	SME
106.	Circuitry	3.6	4.7.10	SME
107.	Identification marking	3.9	4.7.11	Visual

MINOR: None defined

4.7.2 Vibration.- This test shall be conducted in accordance with Method 201A of Standard MIL-STD-202, except that the time period for applied motion in each direction shall be as specified herein. Tests and performance measurements prior to or during vibration are not required for final acceptance. Subsequent to vibration, the unit shall be subjected to all the tests in this specification.

4.7.3 Operating temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" or raised beyond condition "G". When the unit is stabilized at each indicated temperature, it shall be subjected to the tests specified in 4.7.6 to 4.7.10 inclusive and shall meet the requirements specified.

4.7.4 Storage temperature.- This test shall be conducted in accordance with Procedure II, condition "B" and "H" of Specification MIL-F-13926. Upon completion of the storage temperature cycling, the unit shall be subjected to tests specified in 4.7.6 to 4.7.10 inclusive and shall meet the requirements specified. This test shall be performed on the production model only.

4.7.5 Operating voltages.- This test shall be accomplished concurrently with the tests specified in 4.7.6 to 4.7.9 inclusive using a power source with an accuracy of plus or minus one-tenth of one percent to supply the 28 volts dc regulated voltages. A power source to supply 2.5 volts, 400 cycles per second instantaneously positive and negative referred to a common neutral (5v centertapped) having an accuracy of plus or minus one-tenth of one percent shall be used, with the referenced meter or its equal connected to the amplifier to indicate that the unit under test is capable of receiving the voltage specified.

4.7.6 Output signal.- With the amplifier in the operating stage, the deflection observed on the meter shall be in accordance with the requirement specified.

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4.7.7 Output phase.- With a 20-millivolt positive signal applied with the reference voltage, observe the output meter for positive deflection.

4.7.8 Sensitivity.- This test shall be accomplished using a variable voltage source to simulate the specified signal input; through the full cycle of the 20-millivolt input signal observe the deflection on the meter to determine compliance with the requirements specified.

4.7.9 Saturation.- When the amplifier is operating at saturation, observe the reading on the meter to determine compliance with the requirements specified.

4.7.10 Circuitry.- An electrical point-to-point measuring check shall be made using standard measuring equipment to assure that the values of the circuits conform to the electrical schematic.

4.7.11 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The amplifier shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619/P8517. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The nullmeter amplifier is a functional plug-in-subassembly of the gun direction computers (see 1.1). The amplifier is a phase-sensitive null-detecting device used to drive a 100-0-100 microamperes dc ammeter in the replot and voltage test circuits.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

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6.4 Definitions.

6.4.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.2 Special testing device (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.4.3 Tactile.- Skillful manipulation or sensing.

6.4.4 Production model.- A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

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